

SALES ANALYSIS DASHBOARD

An end-to-end Sales Analytics project built in **Excel** and **Power BI**, covering data cleaning, data modeling, DAX measures, and interactive dashboard design.

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Role	Aspiring Data Analyst
Tools Used	Microsoft Excel, Power BI, DAX, Power Query
Dataset	Sales_Dashboard_Dataset_Project.xlsx
Tables	Orders (3,000 rows), Customers (280), Products (25), Salespeople (27)

1. Project Overview

This project analyzes sales performance across regions, countries, product categories, and customer segments using a simulated e-commerce sales dataset. The workflow covers cleaning raw transactional data, building a relational data model, writing DAX measures for key business metrics, and designing an interactive dashboard with slicers for Year, Region, Category, and Payment Mode.

Objective

- Clean and structure raw sales data into a star-schema style data model.
- Build reusable DAX measures for core KPIs (Sales, Profit, Orders, Cost, AOV, Customers).
- Visualize sales trends by year, region, country, category, and order status.
- Enable interactive filtering via slicers for business-user exploration.

2. Data Source & Data Model

The dataset is structured as four related tables, connected through a star schema centered on the Orders fact table.

Table	Rows	Key Fields
Orderstbl (Fact)	3,000	OrderID, OrderDate, CustomerID, ProductID, SalespersonID, Quantity, UnitPrice, Discount, Total Sales, COGS, Profit, ShipMode, PaymentMode, OrderStatus
Customerstbl (Dim)	280	CustomerID, CustomerName, City, State, Segment, JoinDate
Productstbl (Dim)	25	ProductID, ProductName, Category, UnitCost, UnitPrice
Salespeopletbl (Dim)	27	SalespersonID, SalespersonName, Region, HireDate

Relationships: Orderstbl[CustomerID] → Customerstbl[CustomerID], Orderstbl[ProductID] → Productstbl[ProductID], Orderstbl[SalespersonID] → Salespeopletbl[SalespersonID] (all one-to-many, single direction).

3. Key DAX Measures

The dashboard's KPI cards are powered by the following DAX measures, written in Power BI's measure editor:

Total Sales

```
Total Sales = SUM(Orderstbl[Total Sales])
```

Total Profit

```
Total Profit = SUM(Orderstbl[Profit])
```

Total Cost

```
Total Cost = SUM(Orderstbl[COGS])
```

Total Orders

```
Total Orders = DISTINCTCOUNT(Orderstbl[OrderID])
```

Total Customers

```
Total Customers = DISTINCTCOUNT(Orderstbl[CustomerID])
```

Avg Order Value

```
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

Profit Margin %

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

4. Dashboard Preview

The dashboard consolidates KPIs, trend analysis, and segment breakdowns onto a single interactive page, with slicers for Year/Month, Region, Category, and Payment Mode.

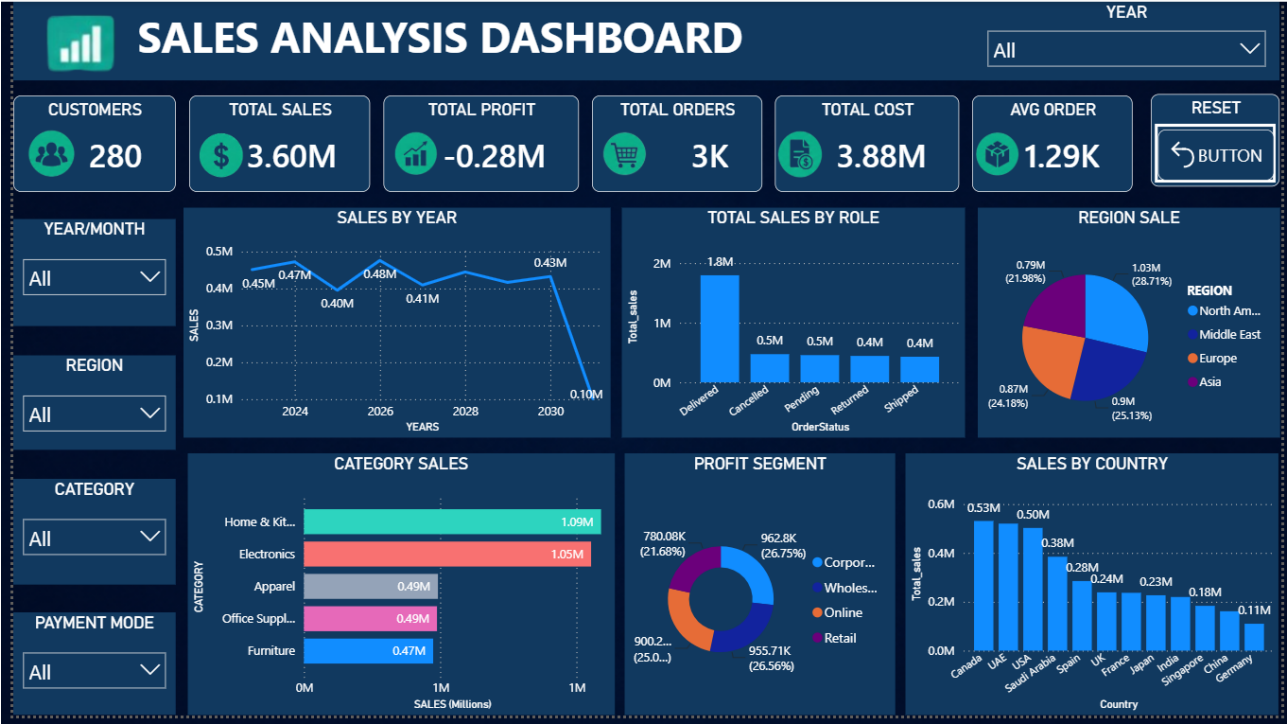


Fig. 1 — Sales Analysis Dashboard (Power BI)

5. Chart-Wise Breakdown & Insights

KPI Cards

Customers: 280 | Total Sales: \$3.60M | Total Profit: -\$0.28M | Total Orders: 3,000 | Total Cost: \$3.88M | Avg Order Value: \$1.29K.

Insight: The business is currently operating at a net loss overall — total cost (\$3.88M) exceeds total sales (\$3.60M), driving the -\$0.28M profit figure. This is the headline insight the dashboard surfaces.

Sales by Year

A line chart tracking Total Sales across the year range, showing fluctuation between roughly \$0.40M and \$0.48M per period before a sharp drop toward the end of the range.

Insight: Sales are broadly stable year-over-year with minor peaks and dips; the steep decline at the final data point likely reflects an incomplete/partial period rather than a genuine downturn, worth flagging when presenting the chart.

Total Sales by Role (Order Status)

A bar chart of Total Sales split by OrderStatus: Delivered (\$1.8M), Cancelled (\$0.5M), Pending (\$0.5M), Returned (\$0.4M), Shipped (\$0.4M).

Insight: Delivered orders make up the majority of recognized sales, but Cancelled and Returned orders together represent a meaningful share of value that never converts to realized revenue — a natural follow-up metric would be a cancellation/return rate KPI.

Region Sale

A pie chart splitting sales across four regions — North America, Middle East, Europe, and Asia — each contributing roughly 22%-29% of total sales.

Insight: Sales are fairly evenly distributed across all four regions, with no single region dominating — suggesting a geographically diversified customer base.

Category Sales

A horizontal bar chart ranking product categories: Home & Kitchen (\$1.09M), Electronics (\$1.05M), Apparel (\$0.49M), Office Supplies (\$0.49M), Furniture (\$0.47M).

Insight: Home & Kitchen and Electronics are the two clear top-performing categories, together contributing close to 60% of category-level sales.

Profit Segment

A donut chart of profit contribution by customer segment: Corporate (\$962.8K, 26.75%), Online (\$955.7K, 26.56%), Wholesale (\$900.2K, ~25%), Retail (\$780.1K, 21.68%).

Insight: Profit contribution is well balanced across segments, with Corporate and Online slightly ahead and Retail trailing — useful for prioritizing segment-specific strategies.

Sales by Country

A bar chart ranking countries by Total Sales, led by Canada (\$0.53M) and UAE (\$0.50M), followed by USA, Saudi Arabia, Spain, UK, France, Japan, India, Singapore, China, and Germany (\$0.11M).

Insight: Sales are concentrated in a handful of top countries (Canada, UAE, USA), while the long tail of countries each contribute smaller, comparable shares — typical of an international but top-heavy customer distribution.

6. Interactivity & Features

- Slicers for Year/Month, Region, Category, and Payment Mode allow dynamic filtering of every visual.
- A Reset button clears all active slicer selections in one click.
- KPI cards, trend line, and breakdown charts update live as filters change.
- Dark themed, single-page layout designed for at-a-glance executive reporting.

7. Skills Demonstrated

- Data modeling (star schema, table relationships)
- DAX measure writing (SUM, DISTINCTCOUNT, DIVIDE)
- Power BI dashboard design (cards, charts, slicers, theming)
- Business/analytical storytelling from KPIs to insights
- Excel-based data preparation and cleaning

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